



VIT CODE APEX 2.0

CODE APEX

HACK THE UPSIDE DOWN

PROBLEM
STATEMENTS

DECRYPT. DESIGN. DEPLOY

VIT CODE APEX TRACK 1 AND TRACK 2 PROBLEM STATEMENTS

TRACK 1: WEB DEVELOPMENT (COMMUDLE)

This track focuses on solving real product engineering challenges at Commudle. The problems range from infrastructure improvements to automation-driven growth systems.

1.1 PROBLEM STATEMENT 1: DOCUMENTATION MIGRATION AND SEARCH SYSTEM

Objective

Commudle currently hosts documentation in Markdown format at <https://documentation.commudle.com>. Build a system that:

- Migrates documentation to a sub-URL such as `commudle.com/documentation`.
- Preserves Markdown structure and formatting.
- Enables full-text search across all Markdown files.
- Supports scalability for future documentation expansion.
- Maintains SEO compatibility and performance standards.

The solution must integrate seamlessly with the existing platform architecture.

1.2 STATEMENT 2: REUSABLE REAL-TIME QUIZ MODULE OBJECTIVE

Objective

Develop a modular, plug-and-play quiz engine that can be embedded anywhere within Commudle. The system should support:

- Quiz creation interface.
- Single-question and multi-question formats.
- Multiple question types (MCQ, subjective, etc.).
- Real-time participant joining.
- Live leaderboard functionality.
- Low-latency real-time updates.
- The UI is aligned with Commudle's design system.

The architecture must be reusable, scalable, and production-ready.

1.3 STATEMENT 3: AUTOMATED SOCIAL MEDIA CONTENT GENERATION FOR FEATURE LAUNCHES

Objective

Commudle frequently launches new features but lacks an automated system for generating promotional content.

Build a system that automatically generates:

- Post captions and launch copy.
- Platform-specific variations (LinkedIn, X, Instagram, etc.).
- Visual creatives (image, carousel, GIF, and short video formats).
- A ready-to-publish content package for each feature release.

The goal is to reduce turnaround time between feature launch and social media deployment.

1.4 PROBLEM STATEMENT 4: AUTOMATED FAQ GENERATION FOR FEATURE TURE PAGES

Objective

Develop a system that automatically generates structured FAQs for newly launched feature pages. The system must:

- Create context-aware FAQs based on feature descriptions.
- Optimize FAQs for SEO.
- Improve discoverability by search engines.
- Enhance AI platform comprehension of the product.

The output should be accurate, structured, and aligned with product intent.

TRACK 1 SUMMARY

Track 1 focuses on product-focused engineering challenges:

- Infrastructure and architecture design.
- Real-time systems.
- Automation pipelines.
- SEO and AI discoverability.

Solutions should be modular, maintainable, and deployment-ready.

TRACK 2: AI AGENTS

This track challenges participants to design and build intelligent AI agents capable of reasoning over structured and unstructured data, extracting insights, enforcing policies, and generating business artifacts. The emphasis is on explainability, modularity, and enterprise-grade scalability.

2.1 PROBLEM STATEMENT 1: DATA DICTIONARY AGENT

Build an AI agent that can analyze relational databases and automatically generate the following:

- Schema understanding (tables, columns, keys, constraints).
- Relationship mapping and ER structure.
- Data quality metrics (nulls, completeness, freshness, consistency).
- AI-generated business context summaries.
- Human-readable data dictionaries.

Participants must handle multi-table relational databases with foreign keys and diverse data types.

Recommended Dataset (Primary)

Olist	Brazilian	E-Commerce	Dataset
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce			
License: CC BY-NC-SA 4.0			

Why it fits:

- 9 interlinked relational tables.
- 100K+ orders.
- Mixed data types (timestamps, categorical, floats, text reviews, geospatial coordinates).
- Real-world schema complexity.
- Data quality variations suitable for profiling.

Alternative Datasets

Bike Store SQL Database
<https://www.kaggle.com/dillonmyrick/bike-store-sample-database> License: CC0 (Public Domain)

Chinook Database <https://github.com/lerocha/chinook-database> License: MIT

Both are suitable for schema extraction and multi-database connector testing.

2.2 PROBLEM STATEMENT 2: BRD (BUSINESS REQUIREMENTS DOCUMENT) AGENT

Objective

Design an AI agent capable of extracting structured business requirements. Documents from noisy communication sources such as

- Emails.
- Meeting transcripts.
- Chat conversations.

The agent must identify and structure the following:

- Functional requirements.
- Non-functional requirements.
- Stakeholders.
- Decisions and approvals.
- Timelines and milestones.
- Feature prioritization.

Recommended Dataset (Primary)

Enron	Email	Dataset
https://www.kaggle.com/datasets/wcukierski/enron-email-dataset		
License: Public Domain		

Why it fits:

- Approximately 500,000 real corporate emails.
- Authentic communication noise.
- Embedded project discussions and decisions.
- Stakeholder hierarchy signals.

Complementary Dataset

AMI	Meeting	Corpus
https://huggingface.co/datasets/knkarthick/AMI		
License: CC BY 4.0		

- 279 meeting transcripts.
- Scenario-based design discussions.
- Includes benchmark summaries.

2.3 PROBLEM STATEMENT 3: DATA POLICY COMPLIANCE AGENT

Objective

Build an AI agent that:

- Ingests free-text policy documents (PDF format).
- Extracts enforceable rules.
- Validates those rules against structured datasets.
- Flags violations with explainable justifications.

The system must provide traceable reasoning for every compliance decision.

Recommended Dataset (Primary):

IBM Transactions for Anti-Money Laundering (AML)

<https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-launderin>

License: CDLA-Sharing-1.0

Why it fits:

- Synthetic but realistic financial transaction data.
- Pre-labeled compliant and violating records.
- Multiple transaction types.
- Suitable for rule-based validation.

Alternative Dataset:

PaySim Financial Dataset

<https://www.kaggle.com/datasets/ealaxi/paysim1> License:

CC BY-SA 4.0

- 6.3 million synthetic transactions.
- Fraud-labeled records.
- Suitable for compliance simulation.